A Project Report

On

Online Shopping Cart

By

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Project Title : Online Shopping Cart

ABSTRACT

An online shopping cart is a software application that enables customers to select and purchase products or services from an e-commerce website.The shopping cart system facilitates a seamless and secure online shopping experience.

Evolution of Shopping Cart Systems : Early shopping carts(1990s) are basic, text-based interfaces, Dynamic shopping carts(2000s) are Database-driven ,interactive, Modern shopping carts(2010s) are Mobile-optimized, socially integrated.

Key features of the online shopping cart include Product Selection and Addition, Quantity Management, Price Calculation, Mode of Payment, Order Summary and Confirmation, Order Tracking.

The system is built using [specify technologies, e.g., JavaScript, HTML, or CSS]and incorporates a user-friendly interface to ensure accessibility for users.

By implementing an Online Shopping Cart, users can enhance Convenience and Easy of Use, 24/7 Shopping Capability, Increased Sales and Revenue, Reduced Operational Costs, enhanced Customer Experience.

SYSTEM ANALYSIS

1. Introduction

a. **Online shopping cart** is a form of electronic commerce whereby consumer directly buy goods or services from a seller over the Internet without an intermediary services. Online shopping is the process whereby consumers directly but goods or services from a seller in real-time , without an intermediary services, over the Internet. The rapid growth of e-commerce has transformed the way consumers shop, making online transactions a norm. At the heart of every e-commerce platform lies a crucial component: the online shopping cart. This program enables customers to select and manage products, facilitating a seamless and personalized shopping experience.

Overview of the Program:

a. This online shopping cart program is designed to simulate a real-world e-commerce scenario, allowing users to:
b. Browse and select products from a virtual store
c. Add and remove items from their shopping cart
d. View and update cart contents
e. Proceed to checkout and complete the purchase.
f. Objectives:
g. The primary objectives of this program are:

h. To develop a user-friendly and interactive online shopping cart system
i. To demonstrate the integration of fundamental data structures and algorithms in a real-world application
j. To provide a comprehensive understanding of the online shopping cart's functionality and its significance in e-commerce

1. **Existing system:**

Components

User Interface: Web-based interface for customers to interact with the shopping cart.

Product Database: Centralized database storing product information, prices, and inventory levels.

Shopping Cart Software: Handles cart operations, such as adding/removing products, calculating totals, and managing sessions.

Payment Gateway: Integrates with payment processors to facilitate secure transactions.

Order Management System: Manages order processing, tracking, and fulfillment.

Functionality

Product browsing: Customers browse products, view details, and add to cart.

Cart Management: Customers view/edit cart contents, apply coupons, and estimate shipping.

Checkout: Customers enter shipping/billing information, select payment options, and review orders.

Payment Processing: Payment gateways process transactions, handle errors, and notify merchants.

Order Fulfillment: Merchants process orders, update inventory, and ship products.

1. **Advantages**

a. Convenience: Online Shopping Carts allow customers to shop from anywhere at any time, and from any device.
b. Increased Sales: Online Shopping Carts can increase sales by making it easy for customers to purchase products online.
c. Improved Customer Experience: Online Shopping Carts provide a seamless and user-friendly experience for customers, making it easy for them to find and purchase products.

2. **Disadvantages**

a. Security concerns: Online shopping carts are vulnerable to security threats, such as hacking, data breaches, and identity theft.

b. Technical issues: Online shopping carts can experience technical issues, such as websites crashes, and errors during checkout.

c. Product return: Online shopping carts can make it difficult for customers to return products, which can lead to customer dissatisfaction.

5.Proposed System

System Components

i. User Interface: A user-friendly web interface for customers to browse products, add/remove items from cart, and checkout.

ii. Product Database: A database to store product information, including descriptions, prices, images, and inventory levels.
iii. Shopping Cart: A session-based cart that stores selected products and quantities.

6.Modules

The modules are:

i. Product Browsing: Customers can browse products by category, search for specific products, and view product details.
ii. Adding/Removing Products: Customers can add/remove products from their shopping cart.
iii. Cart Management: Customers can view/edit their cart contents, including quantities and subtotal.
iv. Checkout: Customers can proceed to checkout, information and selecting payment options.
v. Order Confirmation: Customers receive an order confirmation email with details on their purchase.

7.System Requirements Specifications(SRS)-

a. Software-

E-commerce platform, database management system, payment gateway integration, and security software.

The below code is compatible with operating systems such as WINDOWS 7,8,10,11(32-bit and 64-bit),LINUX(32 bit and 64 bit) and MACOS starting from macOS (cheetah).

a. Hardware-

Reliable web servers, database servers, and payment gateway infrastructure.

The code occupies 5-6KB, the compilers occupy 100-200MB and the total space including the compliers statements etc, is 500-600 MB.

RAM occupies about 16-32KB.

8.Design System:

1.Products Table:

Name

Price

Description

Category

Brand

2. Shopping Cart Table:

Cart

Product

Quantity

Subtotal

3. Orders Table:

Order

Total

Implementation

Code:

```
#include<stdio.h>
#include<string.h>
int main()
{
 static int totalCost;
 int i, j, choice, c, a[9], cost[9];
 for(i = 0; i < 9; i++) a[i] = 0;

 char str[100];
 char items[9][100] = {"men casual shirt", "cotton shirt", "slim fit casual shirt",
                       "adidas", "nike", "leecooper",
                       "organza", "venkatagiri silk saree", "handloom saree"};

 // Input customer name
 printf("Please Enter Your Name: ");
 fflush(stdin); // Clears leftover input
 fgets(str, sizeof(str), stdin);
 str[strcspn(str, "\n")] = '\0'; // Remove newline

 printf("Hello %s, Welcome to our Online Shopping.\n", str);

 do
{
  printf("Enter\n1 - shirts\n2 - Shoes\n3 - sarees\n Any other number to exit\n");
  scanf("%d", &choice);
```

```
 switch(choice)
{
  case 1:
 { // Shirts
   int shirts;
   printf("Enter\n1 - men casual shirt - Rs.355\n2 - cotton shirt- Rs.500\n3 - slim fit casual shirt -
Rs.550\nAny other number to exit\n");
   scanf("%d", &shirts);
   cost[0] = 355; cost[1] = 500; cost[2] = 550;


   switch(shirts)
{
    case 1:
 {
     int num;
     printf("You chose men casual shirt for Rs.355. Confirm? (Enter 1 for Yes): ");
     scanf("%d", &num);
     if(num == 1) { a[0]++; totalCost += 355; }
     printf("Your Cost in Cart: %d\n", totalCost);
     break;
    }
    case 2:
 {
     int num;
     printf("You chose cotton shirt for Rs.500. Confirm? (Enter 1 for Yes): ");
     scanf("%d", &num);
     if(num == 1) { a[1]++; totalCost += 500;
}
     printf("Your Cost in Cart: %d\n", totalCost);
     break;
    }
    case 3:
```

```
{
     int num;
     printf("You chose slim fit casual shirt for Rs.550. Confirm? (Enter 1 for Yes): ");
     scanf("%d", &num);
     if(num == 1) { a[2]++; totalCost += 550;
}
     printf("Your Cost in Cart: %d\n", totalCost);
     break;
    }
    default: printf("Exiting shirts section.\n");
   }
   break;
  }


  case 2:
{ // Shoes
   int shoesChoice;
   printf("Enter\n1 - Adidas - Rs.3550\n2 - Nike - Rs.5000\n3 - Leecooper - Rs.2800\nAny other number to
exit\n");
   scanf("%d", &shoesChoice);
   cost[3] = 3550; cost[4] = 5000; cost[5] = 2800;


   switch(shoesChoice)
{
    case 1:
{
     int num;
     printf("You chose Adidas Shoes for Rs.3550. Confirm? (Enter 1 for Yes): ");
     scanf("%d", &num);
     if(num == 1) { a[3]++; totalCost += 3550;
}
     printf("Your Cost in Cart: %d\n", totalCost);
```

```
      break;
     }
     case 2:
{
      int num;
      printf("You chose Nike Shoes for Rs.5000. Confirm? (Enter 1 for Yes): ");
      scanf("%d", &num);
      if(num == 1) { a[4]++; totalCost += 5000;
}
      printf("Your Cost in Cart: %d\n", totalCost);
      break;
     }
     case 3:
{
      int num;
      printf("You chose Leecooper Shoes for Rs.2800. Confirm? (Enter 1 for Yes): ");
      scanf("%d", &num);
      if(num == 1) { a[5]++; totalCost += 2800;
}
      printf("Your Cost in Cart: %d\n", totalCost);
      break;
     }
     default: printf("Exiting shoes section.\n");
    }
    break;
   }


   case 3:
{ // Sarees
    int sarees;
    printf("Enter\n1 - organza - Rs.11000\n2 - venkatagiri silk saree - Rs.9866\n3 - handloom saree -
Rs.12800\nAny other number to exit\n");
```

```
   scanf("%d", &sarees);
   cost[6] = 11000; cost[7] = 9866; cost[8] = 12800;


   switch(sarees)
{
    case 1:
 {
     int num;
     printf("You chose organza for Rs.11000. Confirm? (Enter 1 for Yes): ");
     scanf("%d", &num);
     if(num == 1) { a[6]++; totalCost += 11000;
}
     printf("Your Cost in Cart: %d\n", totalCost);
     break;
    }
    case 2:
{
     int num;
     printf("You chose venkatagiri silk saree for Rs.9866. Confirm? (Enter 1 for Yes): ");
     scanf("%d", &num);
     if(num == 1)
 {
 a[7]++; totalCost += 9866;
}
     printf("Your Cost in Cart: %d\n", totalCost);
     break;
    }
    case 3:
{
     int num;
     printf("You chose handloom saree for Rs.12800. Confirm? (Enter 1 for Yes): ");
     scanf("%d", &num);
```

```
    if(num == 1) { a[8]++; totalCost += 12800;
}
    printf("Your Cost in Cart: %d\n", totalCost);
    break;
   }
   default: printf("Exiting sarees section.\n");
  }
  break;
 }

 default: printf("Please enter a valid choice.\n");
}

printf("Your Final Cost so far: %d\n", totalCost);
printf("Do you wish to continue shopping? (1 for Yes, any other number to Exit): ");
scanf("%d", &c);

} while(c == 1);

printf("Your Final Cost is %d\n", totalCost);
printf("Thanks %s for Choosing Us and Visit us again.\n", str);
return 0;
}
```

Output

1.Test cases-

Output

```
Please Enter Your Name: sanjana
Hello sanjana, Welcome to our Online Shopping.
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
1
Enter
1 - men casual shirt - Rs.355
2 - cotton shirt- Rs.500
3 - slim fit casual shirt - Rs.550
Any other number to exit
1
You chose men casual shirt for Rs.355. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 355
Your Final Cost so far: 355
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 1
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
2
Enter
1 - Adidas - Rs.3550
2 - Nike - Rs.5000
3 - Leecooper - Rs.2800
```

```
1 - Adidas - Rs.3550
2 - Nike - Rs.5000
3 - Leecooper - Rs.2800
Any other number to exit
1
You chose Adidas Shoes for Rs.3550. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 3905
Your Final Cost so far: 3905
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 1
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
3
Enter
1 - organza - Rs.11000
2 - venkatagiri silk saree - Rs.9866
3 - handloom saree - Rs.12800
Any other number to exit
1
You chose organza for Rs.11000. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 14905
Your Final Cost so far: 14905
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 4
Your Final Cost is 14905
Thanks sanjana for Choosing Us and Visit us again.
```

2.Test case

```
Output
Please Enter Your Name: keerthanasree
Hello keerthanasree, Welcome to our Online Shopping.
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
3
Enter
1 - organza - Rs.11000
2 - venkatagiri silk saree - Rs.9866
3 - handloom saree - Rs.12800
Any other number to exit
3
You chose handloom saree for Rs.12800. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 12800
Your Final Cost so far: 12800
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 1
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
1
```

```
Enter
1 - men casual shirt - Rs.355
2 - cotton shirt- Rs.500
3 - slim fit casual shirt - Rs.550
Any other number to exit
2
You chose cotton shirt for Rs.500. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 13300
Your Final Cost so far: 13300
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 2
Your Final Cost is 13300
Thanks keerthanasree for Choosing Us and Visit us again.
```

3.Test case

Output

```
Please Enter Your Name: esha
Hello esha, Welcome to our Online Shopping.
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
2
Enter
1 - Adidas - Rs.3550
2 - Nike - Rs.5000
3 - Leecooper - Rs.2800
Any other number to exit
3
You chose Leecooper Shoes for Rs.2800. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 2800
Your Final Cost so far: 2800
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 3
Your Final Cost is 2800
Thanks esha for Choosing Us and Visit us again.
```

4.Test case

```
Please Enter Your Name: jyothi
Hello jyothi, Welcome to our Online Shopping.
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
3
Enter
1 - organza - Rs.11000
2 - venkatagiri silk saree - Rs.9866
3 - handloom saree - Rs.12800
Any other number to exit
1
You chose organza for Rs.11000. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 11000
Your Final Cost so far: 11000
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 1
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
3
Enter
1 - organza - Rs.11000
2 - venkatagiri silk saree - Rs.9866
3 - handloom saree - Rs.12800
```

```
Any other number to exit
2
You chose venkatagiri silk saree for Rs.9866. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 20866
Your Final Cost so far: 20866
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 1
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
3
Enter
1 - organza - Rs.11000
2 - venkatagiri silk saree - Rs.9866
3 - handloom saree - Rs.12800
Any other number to exit
3
You chose handloom saree for Rs.12800. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 33666
Your Final Cost so far: 33666
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 4
Your Final Cost is 33666
Thanks jyothi for Choosing Us and Visit us again.
```

5.Test case(delect option usage)

Output

```
Please Enter Your Name: Dasharath
Hello Dasharath, Welcome to our Online Shopping.
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
1
Enter
1 - men casual shirt - Rs.355
2 - cotton shirt- Rs.500
3 - slim fit casual shirt - Rs.550
Any other number to exit
1
You chose men casual shirt for Rs.355. Confirm? (Enter 1 for Yes): 2
Your Cost in Cart: 0
Your Final Cost so far: 0
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 1
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
1
Enter
1 - men casual shirt - Rs.355
2 - cotton shirt- Rs.500
3 - slim fit casual shirt - Rs.550
```

```
Any other number to exit
3
You chose slim fit casual shirt for Rs.550. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 550
Your Final Cost so far: 550
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 1
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
2
Enter
1 - Adidas - Rs.3550
2 - Nike - Rs.5000
3 - Leecooper - Rs.2800
Any other number to exit
2
You chose Nike Shoes for Rs.5000. Confirm? (Enter 1 for Yes): 1
Your Cost in Cart: 5550
Your Final Cost so far: 5550
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 6
Your Final Cost is 5550
Thanks Dasharath for Choosing Us and Visit us again.
```

6.Test case(default case)

Output

```
Please Enter Your Name: aryan
Hello aryan, Welcome to our Online Shopping.
Enter
1 - shirts
2 - Shoes
3 - sarees
Any other number to exit
4
Please enter a valid choice.
Your Final Cost so far: 0
Do you wish to continue shopping? (1 for Yes, any other number to Exit): 4
Your Final Cost is 0
Thanks aryan for Choosing Us and Visit us again.
```